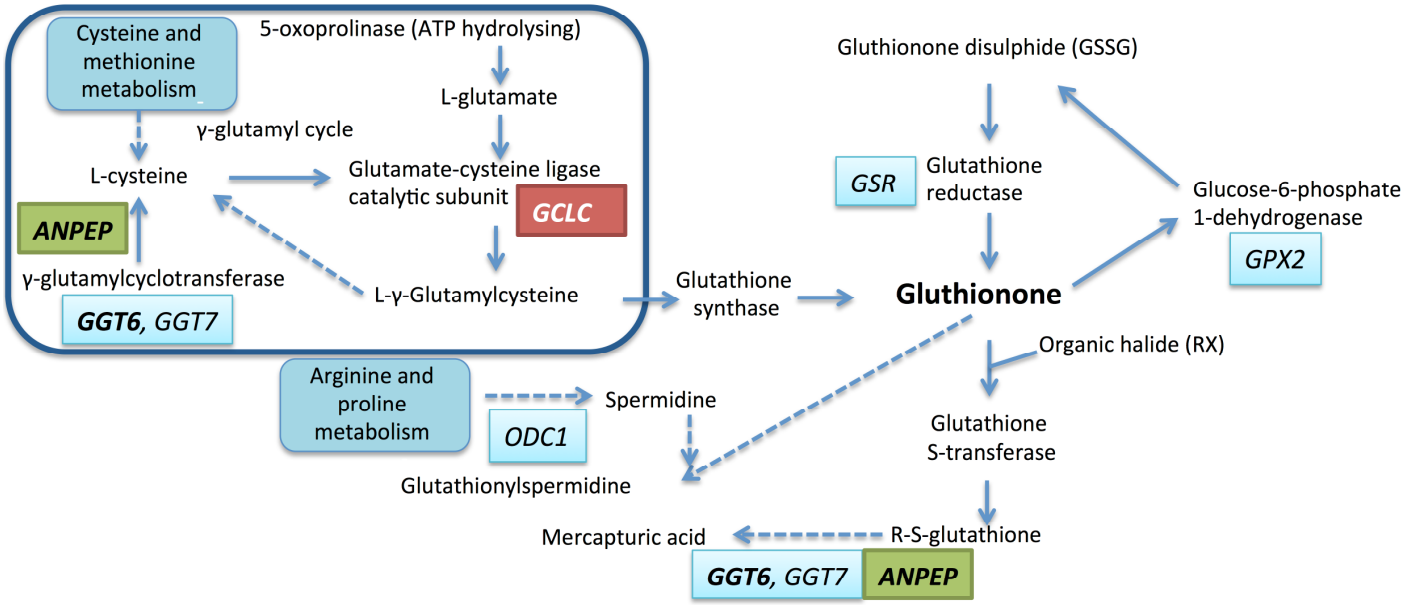
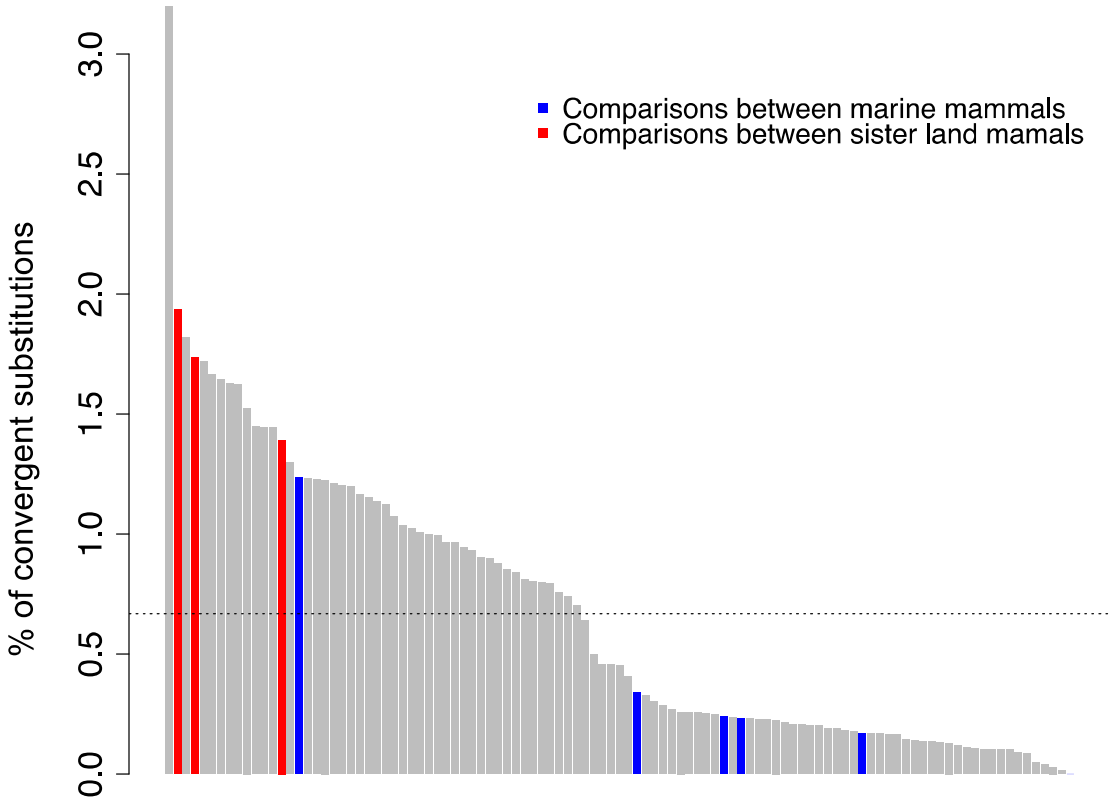


Supplementary Figure 1

Glutathione pathway (KEGG pathway map 00480).

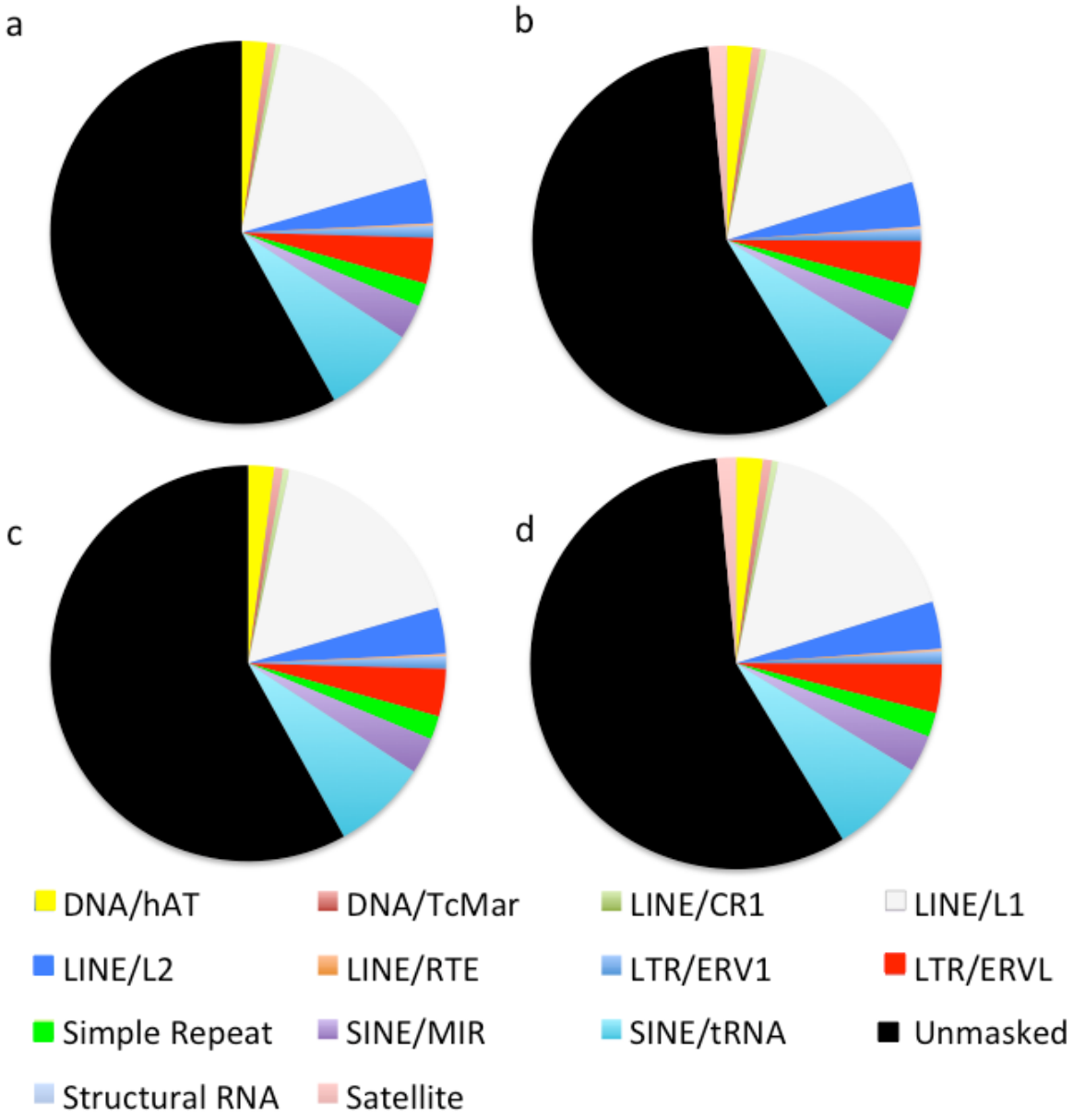
Solid lines indicate direct relationships between enzymes and metabolites, and dashed lines indicate that more than one step is involved in a process. Genes are shown in rectangles. *ANPEP* highlighted in green was one of only five genes found evolving under positive selection along the combined marine mammal branch after correcting for multiple testing. *GCLC* highlighted in red was found to be evolving under positive selection along the combined marine mammal branch and the walrus branch and contained a convergent substitution shared by all three marine mammal lineages. *GGT6* highlighted in bold font was found to be evolving under positive selection along the walrus branch. Yim *et al.*[11] previously found evidence for *GSR* evolving under positive selection in the bottlenose dolphin and experimentally demonstrated the importance of glutathione as an antioxidant in cetacean cells during hypoxic or oxidative stress, e.g., during deep or long dives. This figure is adapted from Figure 3a of Yim *et al.*[11].



Supplementary Figure 2

Percentage of convergent substitutions for all pairwise comparisons among all 14 species in the phylogeny.

The dashed line indicates the average.



Supplementary Figure 3

Pie charts of genome fractions as identified by RepeatMasker.

(**a**)Killer whale.(**b**)Bottlenose dolphin.(**c**)Manatee.(**d**)Walrus.

Supplementary Tables

Supplementary Table 1. Genome assembly statistics for *de novo* genomes sequenced and assembled for this study

Genome Assembly	Bottlenose dolphin Ttru_1.4	Killer whale Oorc_1.1	Walrus Oros_1.0	Manatee TriManLat1.0
Total sequence length	2,551,996,573	2,372,919,877	2,400,150, 193	3,103,808,406
Total assembly gap length	219,573,256	123,337,750	99,898,116	334,708,729
Number of scaffolds	240,901	1,668	3,893	6,323
Scaffold N50	116,261	12,735,091	2,616,778	14,442,683
Number of contigs	554,614	80,100	70,656	166,490
Contig N50	11,819	70,300	89,951	37,750
Sanger sequence coverage	2.5x	n/a	n/a	n/a
Roche 454 sequence coverage	3.5x	n/a	n/a	n/a
Illumina sequence coverage	30x	200x	200x	90x
Genome representation	95%	85%	95%	89%

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Supplementary Table 2. Species for which genome data were included in this study

Common name	Order	Family	Genus	Species
Human	Primates	Hominidae	*Homo*	*sapiens*
Baboon	Primates	Cercopithecidae	*Papio*	*anubis*
Macaque	Primates	Cercopithecidae	*Macaca*	*mulatta*
Marmoset	Primates	Callitrichidae	*Callithrix*	*jacchus*
Mouse	Rodentia	Muridae	*Mus*	*musculus*
Rat	Rodentia	Muridae	*Rattus*	*norvegicus*
Alpaca	Cetartiodactyla	Camelidae	*Vicugna*	*pacos*
Cow	Cetartiodactyla	Bovidae	*Bos*	*taurus*
Dog	Carnivora	Canidae	*Canis*	*familiaris*
Elephant	Proboscidea	Elephantidae	*Loxodonta*	*africana*
Opossum	Didelphimorphia	Didelphidae	*Monodelphis*	*domestica*
Walrus	Carnivora	Pinnipedia	*Odobenus*	*rosmarus**
West Indian manatee	Sirenia	Trichechidae	*Trichechus*	*manatus**
Bottlenose dolphin	Cetartiodactyla	Delphinidae	*Tursiops*	*truncatus***
Killer whale	Cetartiodactyla	Delphinidae	*Orcinus*	*orca**

*species for which *de novo* genomes were sequenced and assembled for this study.
**species for which additional sequencing was done to increase coverage for this study.
Marine mammal species are coloured red.

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Supplementary Table 4. Summary of common and parallel amino acid substitutions found in the three marine mammal lineages and their terrestrial sister taxa (outgroups): the cow, dog and elephant

	3 marine mammal branches	outgroups
Common substitutions	260	608
Percentage of total substitutions	0.28%	0.52%
Parallel substitutions	44	94
Percentage of total substitutions	0.05%	0.08%

	cetacean and walrus branches	outgroups
Common substitutions	1,708	3,341
Percentage of total substitutions	4.23%	6.07%
Parallel substitutions	805	1,560
Percentage of total substitutions	1.36%	1.93%

	cetacean and manatee branches	outgroups
Common substitutions	2,056	4,071
Percentage of total substitutions	4.00%	5.73%
Parallel substitutions	1,021	1,774
Percentage of total substitutions	1.33%	1.74%

	walrus and manatee branches	outgroups
Common substitutions	3,094	4,283
Percentage of total substitutions	3.91%	4.71%
Parallel substitutions	1,401	1,700
Percentage of total substitutions	1.22%	1.34%

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Supplementary Table 5. Summary of common and parallel amino acid substitutions at different likelihood cutoffs and with ancestral CpG sites removed

No cutoff (original results)		
	3 marine mammal branches	outgroups
Common substitutions	260	610
Percentage of total substitutions	0.28	0.52
Parallel substitutions	44	94
Percentage of total substitutions	0.05	0.08

0.9 cutoff		
	3 marine mammal branches	outgroups
Common substitutions	136	279
Percentage of total substitutions	0.17	0.29
Parallel substitutions	31	47
% of changes which are convergent	0.04	0.05

0.95 cutoff		
	3 marine mammal branches	outgroups
Common substitutions	104	201
Percentage of total substitutions	0.13	0.23
Parallel substitutions	24	37
Percentage of total substitutions	0.03	0.04

0.99 cutoff		
	3 marine mammal branches	outgroups
Common substitutions	42	57
Percentage of total substitutions	0.07	0.10
Parallel substitutions	6	9
Percentage of total substitutions	0.01	0.02

No ancestral CpGs		
	3 marine mammal branches	outgroups
Common substitutions	166	388
Percentage of total substitutions	0.24	0.44
Parallel substitutions	29	54
Percentage of total substitutions	0.04	0.06

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Supplementary Table 6. Positively selected genes which contain parallel amino acid substitution in all three marine mammal lineages

*Nucleotide mutations resulting in non-synonymous changes are highlighted in red bold text. Likelihood values of ancestral state are given for each pair of marine/terrestrial sister taxa

Gene	Convergent amino acid substitution	Cetacean/cow ancestral base *P*-value	Manatee/ elephant ancestral base *P*-value	Walrus/dog ancestral base *P*-value	Underlying nucleotide mutations
MYH7B	K➔Q	0.999	0.999	0.998	**A**AG➔**C**AG
TBC1D15	N➔S	1.0	0.985	0.984	A**A**T➔A**G**T
MGP	L➔I	0.968	0.857	0.957	**C**TC➔**A**TC (cetacean & walrus); **C**TT➔**A**TT (manatee)
SMPX	S➔L	0.998	0.998	0.999	T**C**G➔T**T**G
GCLC	V➔M	0.986	0.949	0.997	**G**TG➔**A**TG (cetacean & manatee); **G**TA➔**A**TG (walrus)
SERPINC	N➔S	0.984	0.745	0.975	A**A**C➔A**G**C (cetacean & walrus), A**A**C➔A**G**T (manatee)
M6PR	N➔S	0.804	0.982	0.974	A**A**T➔A**G**T (cetacean & walrus); A**A**C➔A**G**C (manatee)
S100A9	A➔G	0.996	0.607	0.922	G**C**C➔G**G**C (cetacean); G**C**A➔G**G**G (manatee); G**C**C➔G**G**T (walrus)
IRAK2	D➔E	0.584	0.935	0.900	GA**T**➔GA**A** (cetacean); GA**T**➔GA**G** (manatee & walrus)
CHRM5	R➔Q	0.999	0.982	0.939	C**G**G➔C**A**G (cetacean & manatee); C**G**G➔C**A**A (walrus)
GPR97	S➔R	0.892	0.988	0.702	AG**C**➔AG**A** (cetacean); **A**GT➔**C**GT (manatee); AG**C**➔AG**G** (walrus)
ESD	D➔E	0.783	0.439	0.526	GA**C**➔GA**G** (cetacean); GA**T**➔GA**G** (manatee & walrus)
SIAE	I➔V	0.957	0.937	0.970	**AC**A➔**GT**C (cetacean); **A**TA➔**G**TC (manatee); **AC**C➔**GT**C (walrus)
DUSP27	N➔S	0.984	0.979	0.776	A**A**T➔A**G**T (cetacean); **GA**T➔**AG**T (manatee); A**AG**➔A**GC** (walrus)
C7orf62	S➔N	0.997	0.992	0.996	A**G**T➔A**A**T

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Supplementary Table 8. Functional and deleterious associations of genes with convergent substitutions and under positive selection

Gene	UniProtKB/Swiss-Prot Summary of Gene Function	GeneCard Summary of Disorders / Diseases
Myh7b	Involved in muscle contraction	left ventricular noncompaction 1, with or without congenital heart defects acute monocytic leukemia gallbladder cancer monocytic leukemia myocarditis dilated cardiomyopathy coronary artery disease melanoma leukemia breast cancer
Tbc1d15	Acts as a GTPase activating protein for RAB7A.	thyroiditis
Mgp	Associates with the organic matrix of bone and cartilage. Thought to act as an inhibitor of bone formation.	calcification keutel syndrome hyperphosphatemia vitamin k deficiency aortic valve stenosis renal disease atherosclerosis cardiovascular diseases
Smpx	Plays a role in the regulatory network through which muscle cells coordinate their structural and functional states during growth, adaptation, and repair.	deafness and hereditary hearing loss friedreich ataxia labyrinthitis choroiditis myocardial infarction multiple myeloma neuroblastoma neuronitis
Gclc	Catalytic activity: ATP + L-glutamate + L-cysteine = ADP + phosphate + gamma-L-glutamyl-L-cysteine Enzyme regulation: Feedback inhibition by glutathione	anemia hemolytic prostate carcinoma hepatocellular carcinoma cancer metastasis tumors
Serpinc1	Most important serine protease inhibitor in plasma that	antithrombin iii deficiency

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	regulates the blood coagulation cascade. AT-III inhibits thrombin, matriptase-3/TMPRSS7, as well as factors IXa, Xa and XIa. Its inhibitory activity is greatly enhanced in the presence of heparin	thrombophilia protein s deficiency apc resistance thrombosis protein c deficiency disseminated intravascular coagulation thromboembolism venous thrombosis coagulopathy
M6pr	Transport of phosphorylated lysosomal enzymes from the Golgi complex and the cell surface to lysosomes. Lysosomal enzymes bearing phosphomannosyl residues bind specifically to mannose-6-phosphate receptors in the Golgi apparatus and the resulting receptor-ligand complex is transported to an acidic prelyosomal compartment where the low pH mediates the dissociation of the complex	i-cell disease lysosomal storage diseases pompe disease niemann-pick diseases mps iiia fabry disease enzyme deficiency HIV1 tumors necrosis
S100a9	S100A9 is a calcium- and zinc-binding protein which plays a prominent role in the regulation of inflammatory processes and immune response.	inflammation psoriasis inflammatory bowel diseases rheumatoid arthritis cystic fibrosis spondylarthropathies arthritis juvenile rheumatoid dermatomyositis mixed connective tissue disease arthritis
Irak2	Binds to the IL-1 type I receptor following IL-1 engagement, triggering intracellular signaling cascades leading to transcriptional up-regulation and mRNA stabilization	necrosis tumors
Chrm5	The muscarinic acetylcholine receptor mediates various cellular responses, including inhibition of adenylate cyclase, breakdown of phosphoinositides and modulation of potassium channels through the action of G proteins. Primary transducing effect is Pi turnover	schizophrenia
Gpr97	Orphan receptor	hyperparathyroidism
Esd	Serine hydrolase involved in the detoxification of	retinoblastoma

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	formaldehyde	wilson disease
Siae	Catalyzes the removal of O-acetyl ester groups from position 9 of the parent sialic acid, N-acetylneuraminic acid	autoimmune disease 6 rheumatoid arthritis multiple sclerosis Sjogren syndrome type 1 diabetes mellitus primary biliary cirrhosis Crohn disease rheumatoid arthritis leukemia multiple myeloma
Dusp27	protein tyrosine/serine/threonine phosphatase activity	hypertension
C7orf62	exact function unknown	hypothyroidism atherosclerosis endotheliitis

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Supplementary Table 14. Positions of parallel non-synonymous amino acid substitutions that were found in positively selected genes

*The convergent position is show within a sequence of 10 amino acids to facilitate database searches for these regions

MYH7B	Amino acid position 1,881 in the CDS of NM_020884.3									
cow	A	E	Q	K	K	H	A	E	A	L
dolphin	A	E	Q	K	Q	H	A	E	A	L
killer whale	A	E	Q	K	Q	H	A	E	A	L
minke whale	A	E	Q	K	K	H	A	E	A	L
dog	A	E	Q	K	K	H	A	E	A	L
walrus	A	E	Q	K	Q	H	A	E	A	L
elephant	A	E	Q	K	K	H	A	E	A	L
manatee	A	E	Q	K	Q	H	A	E	A	L

IRAK2	Amino acid position 538 in the CDS of NM_001570.3									
cow	N	S	S	F	D	G	S	Y	S	M
dolphin	N	S	S	F	E	A	S	C	C	M
killer whale	N	S	S	F	E	A	S	C	C	M
minke whale	N	S	S	F	E	A	S	C	S	M
dog	N	S	S	L	D	A	S	Y	S	M
walrus	N	S	S	L	E	A	S	S	S	M
elephant	N	S	S	L	D	A	A	P	S	R
manatee	N	S	S	L	E	A	A	S	S	T

TBC1D15	Amino acid position 28 in the CDS of NM_022771.4									
cow	L	G	K	T	N	D	Q	D	S	L
dolphin	F	G	K	T	S	D	Q	D	S	L
killer whale	F	G	K	T	S	D	Q	D	S	L
minke whale	F	G	K	T	S	D	Q	D	S	L
dog	C	G	K	T	N	D	Q	D	S	L
walrus	C	G	K	T	S	D	Q	D	S	L
elephant	C	G	K	T	N	D	Q	D	G	L
manatee	C	G	K	T	S	D	Q	D	T	L

CHRM5	Amino acid position 258 in the CDS of NM_012125.3									
cow	-	-	C	P	W	P	T	L	T	Q
dolphin	S	C	C	P	Q	P	T	L	A	Q
killer whale	S	C	C	P	Q	P	T	L	A	Q
minke whale	S	C	C	P	Q	P	T	L	A	Q
dog	-	-	C	P	R	P	T	L	A	Q
walrus	-	-	C	P	Q	P	T	L	A	Q
elephant	-	-	C	P	R	P	T	L	A	Q
manatee	-	-	C	P	Q	P	T	L	A	Q

MGP	Amino acid position 93 in the CDS of NM_001190839.1									
cow	P	Q	Y	E	L	N	R	E	A	C
dolphin	P	A	Y	E	I	N	R	E	A	C
killer whale	P	A	Y	E	I	N	R	E	A	C
minke whale	P	A	Y	E	I	N	R	E	A	C
dog	P	A	Y	E	L	N	R	E	A	C
walrus	P	A	Y	E	I	S	R	E	A	C
elephant	P	A	H	E	L	N	R	E	A	C
manatee	P	A	H	E	I	N	R	E	A	C

GPR97	Amino acid position 528 of AY140959.1									
cow	L	Y	C	P	S	Q	S	T	V	T
dolphin	L	Y	F	P	R	Q	S	T	V	T
killer whale	L	Y	F	P	R	Q	S	T	V	T
minke whale	L	Y	F	P	S	Q	S	T	M	T
dog	L	Y	F	P	S	R	G	T	M	S
walrus	L	Y	F	P	R	R	S	A	K	A
elephant	L	Y	F	P	S	Q	S	-	A	S
manatee	L	Y	F	P	R	Q	S	-	A	S

SMPX	Amino acid position 50 in the CDS of NM_014332.2									
cow	**A**	**P**	**P**	**T**	**S**	**D**	**E**	**E**	**K**	**K**
dolphin	**A**	**P**	**P**	**T**	**L**	**D**	**E**	**E**	**K**	**K**
killer whale	**A**	**P**	**P**	**T**	**L**	**D**	**E**	**E**	**K**	**K**
minke whale	**A**	**P**	**P**	**T**	**L**	**D**	**E**	**E**	**K**	**K**
dog	**A**	**P**	**P**	**T**	**S**	**D**	**E**	**E**	**K**	**K**
walrus	**V**	**P**	**P**	**T**	**L**	**D**	**E**	**E**	**K**	**K**
elephant	**A**	**P**	**P**	**T**	**S**	**E**	**E**	**E**	**K**	**K**
manatee	**A**	**P**	**P**	**T**	**L**	**D**	**E**	**E**	**K**	**K**

ESD	Amino acid position 95 in the CDS of NM_001984.1									
cow	**K**	**G**	**E**	**D**	**D**	**S**	**W**	**D**	**F**	**G**
dolphin	**K**	**G**	**E**	**D**	**E**	**T**	**W**	**D**	**F**	**G**
killer whale	**K**	**G**	**E**	**E**	**E**	**T**	**W**	**D**	**F**	**G**
minke whale	**K**	**G**	**E**	**E**	**E**	**T**	**W**	**D**	**F**	**G**
dog	**K**	**G**	**E**	**D**	**D**	**S**	**W**	**D**	**F**	**G**
walrus	**K**	**G**	**E**	**D**	**E**	**S**	**W**	**D**	**F**	**G**
elephant	**K**	**G**	**E**	**D**	**D**	**S**	**W**	**D**	**F**	**G**
manatee	**K**	**G**	**E**	**D**	**E**	**S**	**W**	**D**	**F**	**G**

GCLC	Amino acid position 502 in the CDS of NM_001498.3									
cow	**G**	**G**	**N**	**A**	**V**	**V**	**D**	**G**	**C**	**G**
dolphin	**G**	**G**	**N**	**A**	**M**	**V**	**D**	**G**	**R**	**N**
killer whale	**G**	**G**	**N**	**A**	**M**	**V**	**D**	**G**	**R**	**S**
minke whale	**G**	**G**	**N**	**A**	**M**	**V**	**D**	**G**	**R**	**G**
dog	**G**	**G**	**N**	**T**	**V**	**V**	**D**	**G**	**Y**	**G**
walrus	**G**	**G**	**T**	**A**	**M**	**V**	**D**	**G**	**Y**	**S**
elephant	**G**	**G**	**N**	**A**	**V**	**V**	**D**	**G**	**C**	**G**
manatee	**G**	**G**	**N**	**A**	**M**	**V**	**D**	**G**	**Y**	**G**

SIAE	Amino acid position 468 in the CDS of NM_170601.4									
cow	**L**	**A**	**L**	**N**	**I**	**T**	**S**	**C**	**H**	**D**
dolphin	**L**	**A**	**L**	**N**	**V**	**K**	**S**	**C**	**H**	**N**
killer whale	**L**	**A**	**L**	**N**	**V**	**K**	**S**	**C**	**H**	**N**
minke whale	**L**	**A**	**L**	**N**	**V**	**K**	**S**	**C**	**R**	**D**
dog	**L**	**T**	**L**	**S**	**T**	**G**	**S**	**C**	**H**	**G**
walrus	**L**	**A**	**L**	**N**	**V**	**R**	**S**	**C**	**R**	**G**
elephant	**L**	**A**	**L**	**N**	**I**	**N**	**P**	**C**	**H**	**G**
manatee	**L**	**A**	**L**	**D**	**V**	**C**	**P**	**C**	**H**	**G**

SERPINC1	Amino acid position 430 in the CDS of NP_000479.1									
cow	**S**	**L**	**N**	**S**	**D**	**R**	**V**	**T**	**F**	**K**
dolphin	**S**	**L**	**N**	**P**	**S**	**R**	**V**	**T**	**F**	**K**
killer whale	**S**	**L**	**N**	**P**	**S**	**R**	**V**	**T**	**F**	**K**
minke whale	**S**	**L**	**N**	**P**	**S**	**R**	**V**	**T**	**F**	**K**
dog	**S**	**L**	**N**	**P**	**N**	**R**	**V**	**T**	**F**	**K**
walrus	**S**	**L**	**N**	**P**	**S**	**R**	**V**	**T**	**F**	**K**
elephant	**S**	**L**	**N**	**P**	**N**	**R**	**V**	**T**	**F**	**K**
manatee	**S**	**L**	**N**	**P**	**S**	**R**	**V**	**T**	**F**	**K**

DUSP27	Amino acid position 830 in the CDS of NM_001080426.1									
cow	**L**	**F**	**A**	**D**	**N**	**V**	**D**	**L**	**K**	**E**
dolphin	**L**	**F**	**A**	**D**	**S**	**V**	**D**	**L**	**K**	**E**
killer whale	**L**	**F**	**A**	**D**	**S**	**V**	**D**	**L**	**K**	**E**
minke whale	**L**	**F**	**A**	**D**	**S**	**V**	**D**	**L**	**K**	**E**
dog	**L**	**F**	**A**	**D**	**N**	**V**	**D**	**L**	**K**	**E**
walrus	**L**	**F**	**A**	**D**	**S**	**V**	**D**	**L**	**K**	**E**
elephant	**L**	**F**	**A**	**D**	**D**	**V**	**D**	**L**	**K**	**E**
manatee	**L**	**F**	**A**	**D**	**S**	**V**	**D**	**L**	**K**	**E**

M6PR	Amino acid position 130 in the CDS of NM_001207024.1									
cow	D	E	Y	D	N	H	C	G	R	E
dolphin	D	E	Y	D	S	H	C	G	K	E
killer whale	D	E	Y	D	S	H	C	G	K	E
minke whale	D	E	Y	D	S	H	C	G	K	E
dog	D	E	Y	D	N	H	C	G	M	E
walrus	D	E	Y	D	S	H	C	G	M	E
elephant	D	E	Y	D	N	H	C	G	K	E
manatee	D	E	Y	D	S	H	C	G	K	E

C7orf62	Amino acid position 104 in the CDS of NM_152706.3									
cow	L	E	S	S	S	G	T	F	Y	Q
dolphin	L	E	S	S	N	G	T	L	Y	Q
killer whale	L	E	S	S	N	G	T	L	Y	Q
minke whale	L	E	S	S	N	G	T	L	Y	Q
dog	L	E	T	S	S	G	T	L	Y	R
walrus	L	E	S	S	N	G	T	L	Y	R
elephant	L	E	S	S	S	G	T	L	F	R
manatee	L	E	S	S	N	G	T	L	F	R

S100A9	Amino acid position 84 in the CDS of NM_002965.3									
cow	I	M	L	V	A	R	L	T	V	A
dolphin	S	V	L	V	G	K	L	T	E	A
killer whale	S	V	L	V	G	K	L	T	E	A
minke whale	L	I	L	V	A	G	L	T	E	A
dog	S	I	L	V	A	R	L	T	V	A
walrus	S	I	L	V	G	R	L	T	L	A
elephant	I	V	L	V	A	K	L	T	E	A
Manatee	S	I	L	V	G	Q	L	T	E	A